

ADAL Database Schema Design

AI-Driven Analysis for Law

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Overview

The ADAL database schema is designed to support a comprehensive legal intelligence platform with the following key functionalities:

- User authentication and role-based access control
- Document storage and processing
- OCR results and text extraction
- Legal citation detection and validation
- Precedent search and relevance ranking
- Real-time collaboration
- Judge analytics and case insights

Design Principles

- Scalability:** Schema designed to handle growing legal document repositories

- **Performance:** Optimized for fast search and retrieval operations
 - **Security:** Encrypted sensitive data with proper access controls
 - **Flexibility:** Extensible design for future feature additions
 - **Compliance:** Adheres to legal data retention and privacy requirements
-

Database Technology Stack

Primary Database: PostgreSQL 14+

- **Rationale:** ACID compliance, JSON support, full-text search, robust indexing
- **Extensions:** pg_trgm (trigram matching), pg_stat_statements (performance monitoring)

Search Engine: Elasticsearch 8.0+

- **Purpose:** Full-text search, semantic search, legal precedent ranking
- **Integration:** Synchronized with PostgreSQL via background jobs

Caching Layer: Redis 7.0+

- **Purpose:** Session storage, query result caching, real-time collaboration
 - **Data Types:** Strings, Hashes, Sets, Sorted Sets, Streams
-

Schema Architecture

Database Structure



Core Tables

1. Authentication & User Management

users

sql

```
CREATE TABLE users (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  email VARCHAR(255) UNIQUE NOT NULL,  
  password_hash VARCHAR(255) NOT NULL,  
  first_name VARCHAR(100) NOT NULL,  
  last_name VARCHAR(100) NOT NULL,  
  role user_role_enum NOT NULL DEFAULT 'user',  
  organization_id UUID REFERENCES organizations(id),  
  is_active BOOLEAN DEFAULT TRUE,  
  is_verified BOOLEAN DEFAULT FALSE,  
  profile_image_url TEXT,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  last_login_at TIMESTAMP WITH TIME ZONE,  
  password_reset_token VARCHAR(255),  
  password_reset_expires TIMESTAMP WITH TIME ZONE,  
  verification_token VARCHAR(255)  
);  
  
CREATE TYPE user_role_enum AS ENUM ('admin', 'lawyer', 'researcher', 'student', 'user');
```

organizations

sql

```
CREATE TABLE organizations (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  name VARCHAR(255) NOT NULL,  
  type org_type_enum NOT NULL,  
  address TEXT,  
  phone VARCHAR(20),  
  email VARCHAR(255),  
  website VARCHAR(255),  
  license_number VARCHAR(100),  
  is_active BOOLEAN DEFAULT TRUE,  
  subscription_plan plan_enum DEFAULT 'basic',  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);  
  
CREATE TYPE org_type_enum AS ENUM ('law_firm', 'court', 'university', 'government', 'individual');  
CREATE TYPE plan_enum AS ENUM ('basic', 'professional', 'enterprise');
```

user_sessions

sql

```
CREATE TABLE user_sessions (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,  
  token_hash VARCHAR(255) UNIQUE NOT NULL,  
  device_info JSONB,  
  ip_address INET,  
  user_agent TEXT,  
  expires_at TIMESTAMP WITH TIME ZONE NOT NULL,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  last_activity TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

2. Document Management

documents

sql

```

CREATE TABLE documents (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES users(id),
  organization_id UUID REFERENCES organizations(id),
  title VARCHAR(500) NOT NULL,
  description TEXT,
  document_type doc_type_enum NOT NULL,
  language language_enum NOT NULL DEFAULT 'english',
  file_path TEXT NOT NULL,
  file_name VARCHAR(255) NOT NULL,
  file_size BIGINT NOT NULL,
  mime_type VARCHAR(100) NOT NULL,
  file_hash VARCHAR(64) NOT NULL, -- SHA-256 hash for integrity
  storage_provider storage_enum DEFAULT 'aws_s3',
  is_processed BOOLEAN DEFAULT FALSE,
  processing_status status_enum DEFAULT 'pending',
  processing_started_at TIMESTAMP WITH TIME ZONE,
  processing_completed_at TIMESTAMP WITH TIME ZONE,
  processing_error TEXT,
  is_public BOOLEAN DEFAULT FALSE,
  is_archived BOOLEAN DEFAULT FALSE,
  folder_id UUID REFERENCES document_folders(id),
  tags TEXT[],
  metadata JSONB,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
  version_number INTEGER DEFAULT 1
);

```

```

CREATE TYPE doc_type_enum AS ENUM (
  'contract', 'judgment', 'petition', 'appeal', 'ordinance',
  'constitution', 'regulation', 'case_law', 'legal_opinion',
  'brief', 'memo', 'agreement', 'other'
);

```

```

CREATE TYPE language_enum AS ENUM ('english', 'urdu', 'mixed');
CREATE TYPE storage_enum AS ENUM ('aws_s3', 'azure_blob', 'local');
CREATE TYPE status_enum AS ENUM ('pending', 'processing', 'completed', 'failed', 'cancelled');

```

document_folders

```
sql
```

```
CREATE TABLE document_folders (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID NOT NULL REFERENCES users(id),  
  organization_id UUID REFERENCES organizations(id),  
  name VARCHAR(255) NOT NULL,  
  description TEXT,  
  parent_folder_id UUID REFERENCES document_folders(id),  
  path TEXT, -- Materialized path for efficient querying  
  is_shared BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

document_versions

sql

```
CREATE TABLE document_versions (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE,  
  version_number INTEGER NOT NULL,  
  file_path TEXT NOT NULL,  
  file_hash VARCHAR(64) NOT NULL,  
  changes_summary TEXT,  
  created_by UUID NOT NULL REFERENCES users(id),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  UNIQUE(document_id, version_number)  
);
```

3. OCR & Text Processing

ocr_results

sql

```
CREATE TABLE ocr_results (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE,
  page_number INTEGER NOT NULL,
  extracted_text TEXT NOT NULL,
  confidence_score DECIMAL(5,4), -- Overall confidence (0.0000 to 1.0000)
  language_detected language_enum,
  processing_time_ms INTEGER,
  bounding_boxes JSONB, -- Store word/line level bounding boxes
  preprocessing_applied TEXT[], -- List of preprocessing steps
  ocr_engine VARCHAR(50) DEFAULT 'tesseract',
  engine_version VARCHAR(20),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

text_corrections

```
sql

CREATE TABLE text_corrections (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  ocr_result_id UUID NOT NULL REFERENCES ocr_results(id) ON DELETE CASCADE,
  original_text TEXT NOT NULL,
  corrected_text TEXT NOT NULL,
  correction_type correction_type_enum NOT NULL,
  confidence_score DECIMAL(5,4),
  position_start INTEGER NOT NULL,
  position_end INTEGER NOT NULL,
  corrected_by UUID REFERENCES users(id),
  is_verified BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE TYPE correction_type_enum AS ENUM ('manual', 'auto_spell', 'auto_grammar', 'ai_suggestion');
```

4. Document Structure & Parsing

document_structures

```
sql
```

```

CREATE TABLE document_structures (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE,
  structure_type structure_type_enum NOT NULL,
  section_number VARCHAR(50),
  title VARCHAR(500),
  content TEXT NOT NULL,
  parent_section_id UUID REFERENCES document_structures(id),
  level INTEGER NOT NULL DEFAULT 1,
  order_index INTEGER NOT NULL,
  start_page INTEGER,
  end_page INTEGER,
  confidence_score DECIMAL(5,4),
  parsing_method VARCHAR(50),
  metadata JSONB,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE TYPE structure_type_enum AS ENUM (
  'title', 'chapter', 'section', 'subsection', 'article',
  'clause', 'paragraph', 'list_item', 'footnote', 'header', 'footer'
);

```

5. Legal Citations

citations

```

sql

```



```

CREATE TABLE citations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE,
  citation_text TEXT NOT NULL,
  citation_type citation_type_enum NOT NULL,
  case_name VARCHAR(500),
  court VARCHAR(255),
  year INTEGER,
  volume VARCHAR(50),
  reporter VARCHAR(100),
  page VARCHAR(50),
  jurisdiction VARCHAR(100),
  citation_format citation_format_enum,
  position_start INTEGER NOT NULL,
  position_end INTEGER NOT NULL,
  context_before TEXT,
  context_after TEXT,
  is_validated BOOLEAN DEFAULT FALSE,
  validation_status validation_status_enum DEFAULT 'pending',
  validation_source VARCHAR(255),
  confidence_score DECIMAL(5,4),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
  validated_at TIMESTAMP WITH TIME ZONE
);

CREATE TYPE citation_type_enum AS ENUM (
  'case_law', 'statute', 'regulation', 'constitutional',
  'international', 'journal', 'book', 'other'
);

CREATE TYPE citation_format_enum AS ENUM ('pakistani', 'indian', 'uk', 'us', 'international');
CREATE TYPE validation_status_enum AS ENUM ('pending', 'valid', 'invalid', 'uncertain', 'manual_review');

```

legal_databases

sql

```
CREATE TABLE legal_databases (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  name VARCHAR(255) NOT NULL,  
  description TEXT,  
  jurisdiction VARCHAR(100),  
  data_source VARCHAR(255),  
  api_endpoint TEXT,  
  last_updated TIMESTAMP WITH TIME ZONE,  
  is_active BOOLEAN DEFAULT TRUE,  
  access_credentials JSONB, -- Encrypted API keys/credentials  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

6. Legal Precedents & Search

precedents

sql

```
CREATE TABLE precedents (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  case_name VARCHAR(500) NOT NULL,  
  case_number VARCHAR(100),  
  court VARCHAR(255) NOT NULL,  
  judge_names TEXT[],  
  decision_date DATE,  
  jurisdiction VARCHAR(100),  
  case_type VARCHAR(100),  
  legal_areas TEXT[], -- Areas of law (contract, criminal, etc.)  
  keywords TEXT[],  
  summary TEXT,  
  full_text TEXT,  
  citation VARCHAR(500),  
  parties_plaintiff TEXT[],  
  parties_defendant TEXT[],  
  outcome VARCHAR(100),  
  precedent_value precedent_value_enum,  
  language language_enum DEFAULT 'english',  
  source_database VARCHAR(100),  
  external_id VARCHAR(255),  
  embedding_vector VECTOR(384), -- For semantic search  
  is_landmark BOOLEAN DEFAULT FALSE,  
  relevance_score DECIMAL(5,4),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);  
  
CREATE TYPE precedent_value_enum AS ENUM ('binding', 'persuasive', 'informative');
```

search_queries

sql

```
CREATE TABLE search_queries (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES users(id),
  query_text TEXT NOT NULL,
  search_type search_type_enum NOT NULL,
  filters JSONB,
  results_count INTEGER,
  search_time_ms INTEGER,
  clicked_results UUID[],
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

CREATE TYPE search_type_enum AS ENUM ('keyword', 'semantic', 'citation', 'similar_cases');
```

7. Collaboration & Real-time Features

collaborations

sql

```
CREATE TABLE collaborations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE,
  owner_id UUID NOT NULL REFERENCES users(id),
  collaborator_id UUID NOT NULL REFERENCES users(id),
  permission_level permission_enum NOT NULL,
  invited_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
  accepted_at TIMESTAMP WITH TIME ZONE,
  status collab_status_enum DEFAULT 'pending',
  invitation_token VARCHAR(255),
  expires_at TIMESTAMP WITH TIME ZONE,
  UNIQUE(document_id, collaborator_id)
);

CREATE TYPE permission_enum AS ENUM ('read', 'comment', 'edit', 'admin');
CREATE TYPE collab_status_enum AS ENUM ('pending', 'accepted', 'declined', 'revoked');
```

document_comments

sql

```
CREATE TABLE document_comments (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE,
  user_id UUID NOT NULL REFERENCES users(id),
  content TEXT NOT NULL,
  position_start INTEGER,
  position_end INTEGER,
  page_number INTEGER,
  thread_id UUID REFERENCES document_comments(id),
  is_resolved BOOLEAN DEFAULT FALSE,
  resolved_by UUID REFERENCES users(id),
  resolved_at TIMESTAMP WITH TIME ZONE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

document_edits

sql

```
CREATE TABLE document_edits (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  document_id UUID NOT NULL REFERENCES documents(id) ON DELETE CASCADE,
  user_id UUID NOT NULL REFERENCES users(id),
  edit_type edit_type_enum NOT NULL,
  position_start INTEGER,
  position_end INTEGER,
  old_content TEXT,
  new_content TEXT,
  timestamp TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
  session_id VARCHAR(255)
);
```

```
CREATE TYPE edit_type_enum AS ENUM ('insert', 'delete', 'replace', 'format');
```

8. Analytics & Reporting

judges

sql

```
CREATE TABLE judges (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  name VARCHAR(255) NOT NULL,  
  title VARCHAR(100),  
  court VARCHAR(255),  
  jurisdiction VARCHAR(100),  
  appointment_date DATE,  
  retirement_date DATE,  
  specializations TEXT[],  
  bio TEXT,  
  photo_url TEXT,  
  is_active BOOLEAN DEFAULT TRUE,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

case_analytics

sql

```
CREATE TABLE case_analytics (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  precedent_id UUID NOT NULL REFERENCES precedents(id),  
  judge_id UUID REFERENCES judges(id),  
  case_duration_days INTEGER,  
  case_complexity_score DECIMAL(5,2),  
  decision_type decision_type_enum,  
  appeal_outcome appeal_outcome_enum,  
  legal_principles TEXT[],  
  cited_precedents UUID[],  
  statistical_data JSONB,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);  
  
CREATE TYPE decision_type_enum AS ENUM ('unanimous', 'majority', 'split', 'dissent');  
CREATE TYPE appeal_outcome_enum AS ENUM ('upheld', 'overturned', 'remanded', 'modified');
```

9. System Configuration

system_settings

sql

```
CREATE TABLE system_settings (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  key VARCHAR(100) UNIQUE NOT NULL,  
  value JSONB NOT NULL,  
  description TEXT,  
  is_encrypted BOOLEAN DEFAULT FALSE,  
  last_modified_by UUID REFERENCES users(id),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

audit_logs

```
sql  
  
CREATE TABLE audit_logs (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID REFERENCES users(id),  
  action VARCHAR(100) NOT NULL,  
  resource_type VARCHAR(50) NOT NULL,  
  resource_id UUID,  
  old_values JSONB,  
  new_values JSONB,  
  ip_address INET,  
  user_agent TEXT,  
  session_id VARCHAR(255),  
  timestamp TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

Relationships & Constraints

Primary Relationships

1. **Users** ↔ **Organizations**: Many-to-One (users belong to organizations)
2. **Users** ↔ **Documents**: One-to-Many (users can own multiple documents)
3. **Documents** ↔ **OCR Results**: One-to-Many (documents can have multiple pages)
4. **Documents** ↔ **Citations**: One-to-Many (documents contain multiple citations)
5. **Documents** ↔ **Collaborations**: One-to-Many (documents can have multiple collaborators)
6. **Precedents** ↔ **Judges**: Many-to-Many (cases can have multiple judges)

Constraint Definitions

```
sql
```

-- Ensure valid email formats

```
ALTER TABLE users ADD CONSTRAINT valid_email  
CHECK (email ~* '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$');
```

-- Ensure positive file sizes

```
ALTER TABLE documents ADD CONSTRAINT positive_file_size  
CHECK (file_size > 0);
```

-- Ensure valid confidence scores

```
ALTER TABLE ocr_results ADD CONSTRAINT valid_confidence  
CHECK (confidence_score >= 0 AND confidence_score <= 1);
```

-- Ensure valid year ranges for cases

```
ALTER TABLE precedents ADD CONSTRAINT valid_year  
CHECK (EXTRACT(YEAR FROM decision_date) >= 1850 AND  
EXTRACT(YEAR FROM decision_date) <= EXTRACT(YEAR FROM NOW()));
```

-- Ensure document structure hierarchy

```
ALTER TABLE document_structures ADD CONSTRAINT valid_hierarchy  
CHECK (level > 0 AND level <= 10);
```

Indexing Strategy

Primary Indexes

sql

-- User authentication & lookup

CREATE INDEX idx_users_email ON users(email);

CREATE INDEX idx_users_organization ON users(organization_id) WHERE is_active = TRUE;

CREATE INDEX idx_user_sessions_token ON user_sessions(token_hash);

CREATE INDEX idx_user_sessions_user ON user_sessions(user_id, expires_at);

-- Document management

CREATE INDEX idx_documents_user ON documents(user_id, created_at DESC);

CREATE INDEX idx_documents_type ON documents(document_type, is_archived);

CREATE INDEX idx_documents_processing ON documents(processing_status, processing_started_at);

CREATE INDEX idx_documents_tags ON documents USING GIN(tags);

CREATE INDEX idx_documents_metadata ON documents USING GIN(metadata);

-- Full-text search

CREATE INDEX idx_documents_title_search ON documents USING GIN(to_tsvector('english', title));

CREATE INDEX idx_ocr_text_search ON ocr_results USING GIN(to_tsvector('english', extracted_text));

-- OCR and text processing

CREATE INDEX idx_ocr_document_page ON ocr_results(document_id, page_number);

CREATE INDEX idx_ocr_confidence ON ocr_results(confidence_score DESC);

-- Citations and legal references

CREATE INDEX idx_citations_document ON citations(document_id, citation_type);

CREATE INDEX idx_citations_validation ON citations(validation_status, is_validated);

CREATE INDEX idx_citations_case_name ON citations USING GIN(to_tsvector('english', case_name));

-- Precedent search

CREATE INDEX idx_precedents_court_date ON precedents(court, decision_date DESC);

CREATE INDEX idx_precedents_keywords ON precedents USING GIN(keywords);

CREATE INDEX idx_precedents_legal_areas ON precedents USING GIN(legal_areas);

CREATE INDEX idx_precedents_fulltext ON precedents USING GIN(to_tsvector('english', full_text));

-- Collaboration

CREATE INDEX idx_collaborations_user ON collaborations(collaborator_id, status);

CREATE INDEX idx_collaborations_document ON collaborations(document_id, permission_level);

CREATE INDEX idx_comments_document ON document_comments(document_id, created_at DESC);

-- Analytics and performance

CREATE INDEX idx_search_queries_user ON search_queries(user_id, created_at DESC);

CREATE INDEX idx_audit_logs_user_action ON audit_logs(user_id, action, timestamp DESC);

CREATE INDEX idx_audit_logs_resource ON audit_logs(resource_type, resource_id, timestamp DESC);

Composite Indexes for Complex Queries

sql

-- Document search with filters

```
CREATE INDEX idx_documents_complex_search ON documents(  
  user_id, document_type, language, is_archived, created_at DESC  
);
```

-- Citation validation workflow

```
CREATE INDEX idx_citations_validation_workflow ON citations(  
  document_id, validation_status, citation_type, created_at  
);
```

-- Precedent relevance ranking

```
CREATE INDEX idx_precedents_relevance ON precedents(  
  jurisdiction, case_type, precedent_value, decision_date DESC  
);
```

Migration Strategy

Phase 1: Core Schema (Weeks 1-2)

1. User authentication and basic document management
2. Essential tables: users, organizations, user_sessions, documents
3. Basic indexing for authentication and document operations

Phase 2: Document Processing (Weeks 3-4)

1. OCR and text processing tables
2. Document structure and parsing
3. Full-text search indexes

Phase 3: Legal Intelligence (Weeks 5-8)

1. Citations and legal databases
2. Precedent management
3. Search and analytics tables

Phase 4: Collaboration (Weeks 9-10)

1. Real-time collaboration features
2. Comments and edit tracking
3. Advanced analytics

Phase 5: Optimization (Weeks 11-14)

1. Performance tuning
2. Additional indexes based on usage patterns
3. Data archival strategies

Migration Scripts Structure

```
sql

-- migrations/001_initial_schema.sql
-- migrations/002_add_ocr_tables.sql
-- migrations/003_add_citations.sql
-- migrations/004_add_collaboration.sql
-- migrations/005_add_analytics.sql
```

Security Considerations

Data Encryption

- **At Rest:** Sensitive fields encrypted using PostgreSQL's pgcrypto extension
- **In Transit:** All connections use TLS 1.3
- **Application Level:** Additional encryption for PII and legal content

Access Control

```
sql

-- Row-level security for multi-tenancy
ALTER TABLE documents ENABLE ROW LEVEL SECURITY;

CREATE POLICY document_access_policy ON documents
FOR ALL TO authenticated_users
USING (user_id = current_user_id() OR
      id IN (SELECT document_id FROM collaborations
            WHERE collaborator_id = current_user_id()
            AND status = 'accepted'));

-- Similar policies for other sensitive tables
ALTER TABLE ocr_results ENABLE ROW LEVEL SECURITY;
ALTER TABLE citations ENABLE ROW LEVEL SECURITY;
ALTER TABLE collaborations ENABLE ROW LEVEL SECURITY;
```

Audit Trail

- All modifications logged in audit_logs table
- Automatic triggers for sensitive operations
- Retention policy for compliance requirements

Data Anonymization

```
sql

-- Function to anonymize user data for development/testing
CREATE OR REPLACE FUNCTION anonymize_user_data()
RETURNS void AS $$
BEGIN
    UPDATE users SET
        email = 'user_' || id || '@example.com',
        first_name = 'Test',
        last_name = 'User' || substring(id::text, 1, 8)
    WHERE role != 'admin';
END;
$$ LANGUAGE plpgsql;
```

Performance Optimization

Query Optimization Strategies

1. Partition Large Tables

```
sql

-- Partition audit logs by month
CREATE TABLE audit_logs_y2025m01 PARTITION OF audit_logs
    FOR VALUES FROM ('2025-01-01') TO ('2025-02-01');
```

2. Materialized Views for Analytics

```
sql
```

```

CREATE MATERIALIZED VIEW user_document_stats AS
SELECT
  u.id as user_id,
  u.first_name || ' ' || u.last_name as full_name,
  COUNT(d.id) as document_count,
  AVG(o.confidence_score) as avg_ocr_confidence,
  COUNT(c.id) as citation_count
FROM users u
LEFT JOIN documents d ON u.id = d.user_id
LEFT JOIN ocr_results o ON d.id = o.document_id
LEFT JOIN citations c ON d.id = c.document_id
GROUP BY u.id, u.first_name, u.last_name;

-- Refresh strategy
CREATE OR REPLACE FUNCTION refresh_user_stats()
RETURNS void AS $$
BEGIN
  REFRESH MATERIALIZED VIEW CONCURRENTLY user_document_stats;
END;
$$ LANGUAGE plpgsql;

```

3. Database Connection Pooling

- Use PgBouncer for connection pooling
- Configure appropriate pool sizes for different environments

Monitoring Queries

sql

```
-- Query performance monitoring
```

```
SELECT
```

```
    query,  
    calls,  
    total_time,  
    mean_time,  
    rows
```

```
FROM pg_stat_statements
```

```
ORDER BY total_time DESC
```

```
LIMIT 10;
```

```
-- Index usage analysis
```

```
SELECT
```

```
    schemaname,  
    tablename,  
    indexname,  
    idx_scan,  
    idx_tup_read,  
    idx_tup_fetch
```

```
FROM pg_stat_user_indexes
```

```
ORDER BY idx_scan DESC;
```

This database schema provides a robust foundation for the ADAL legal intelligence platform, supporting all planned features while maintaining performance, security, and scalability requirements.